

**A Multimodal Edge-AI Framework for Early Disease Risk
Prediction and Severity-based Resource Optimization in
Cavendish Banana Plantations in Sri Lanka**

Multimodal Disease Progression and Severity Forecasting

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Various diseases of the leaves and trunk attack Cavendish banana plantations, including Black Sigatoka, Yellow Sigatoka, Cordana leaf spot, and Banana Bunchy Top Virus (BBTV), which cause a gradual loss of leaf area, fruit production and quality. Manual severity assessment, such as the Stover–Gauhl visual scale, is still the method of choice, is subjective, time consuming, and relies on the individual's experience, and does not offer any indication of the process of infection, whether it is going up, down or staying stable. The current computational studies have mostly focused on disease classification, and when severity is studied, the information content of images, environment, and time are generally treated separately in the same study.

This proposal describes the third component of a proposed four-component framework for a multimodal approach to Cavendish banana disease severity assessment and short-term prediction of disease progression. The component determines disease classification via Component 1 and an environmental disease-risk score via Component 2, while also deriving image-derived leaf and lesion area quantification from the U-Net model, and integrates them into a continuous score of disease severity, as well as an interpretable category (Low, Moderate, High, Critical). The classification of progression is based on repeated observations of the same plant, and the investigation of short term forecasting with a Gated Recurrent Unit model, compared to simpler baselines.

The proposed concept of the fusion between Impact, Urgency and Scope is assumed as an unproven and exploratory framework, and assessed based on severity rankings assigned by experts, in collaboration with a selected private Cavendish banana plantation. The methodology includes image segmentation, classification with XGBoost and temporal modelling with GRU, which were evaluated with segmentation, classification, regression and forecasting metrics. The main contribution is the association of image, environmental and temporal modalities for the banana severity assessment, in addition to practical proxies for BBTV and Cordana severity for which there is no validated severity standard.

Keywords: Disease Severity Assessment, Banana Disease Progression, Image Segmentation, Multimodal Fusion, Severity Forecasting.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BBTV	Banana Bunchy Top Virus
CNN	Convolutional Neural Network
CBAM	Convolutional Block Attention Module
DSI	Disease Severity Index
GRU	Gated Recurrent Unit
IoU	Intersection over Union
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
R ²	Coefficient of Determination
RMSE	Root Mean Squared Error
SADs	Standard Area Diagram Sets
SVM	Support Vector Machine
SLIIT	Sri Lanka Institute of Information Technology
XGBoost	Extreme Gradient Boosting

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1. INTRODUCTION

1.1 Background and Context

In Sri Lanka, banana is the most important smallholder and commercial crop, and its Cavendish plantations are under threat from foliar and systemic diseases, such as Black Sigatoka (*Pseudocercospora fijiensis*), Yellow Sigatoka (*Pseudocercospora musicola*), Cordana leaf spot (*Neocordana musae*), and Banana Bunchy Top Virus (BBTV), which reduce photosynthetic leaf area, fruit yield, and fruit quality [7]. Visual field inspection is generally the only way most small/medium-scale growers can detect disease which is usually done at a stage where disease symptoms have already fully manifested, thus delaying treatment and leading to higher use of agrochemicals in the field. The overall group research focuses on this challenge within the framework of a multimodal edge-AI decision support system with four components: early disease detection (Component 1), environmental disease-risk prediction (Component 2), disease severity assessment and forecasting of disease progression (Component 3 – this proposal), and severity-based resource optimization (Component 4).

Name of the disease is not enough to give any specific information to take any action of plantation management. Agricultural officers need more information, including the severity of the current infection, what percentage of the leaf area is being affected, if current environmental conditions are conducive to further spread and if the infection is increasing, remaining at the same level or decreasing compared to previous observations. The questions relate, respectively, to the disease impact, disease scope, environmental urgency, and the temporal progression — the four dimensions that are addressed by Component 3.

Most banana disease research is focused on image-based disease classification, which involves determining which disease is present, using convolutional neural network and object-detection architectures and datasets created primarily in Tanzania and other EA countries [10],[11],[23]. Much less research focuses on the severity of the disease: the severity of a known infection and changes in severity over time. Where severity is measured, it is most often done manually with ordinal visual scales, most commonly the Stover–Gauhl scale for Black Sigatoka, which later were developed into disease-specific Standard Area Diagram Sets (SADs) [3],[4],[5]. The manual methods demand trained assessors, are time consuming at plantation level and are subjectively determined, with inter-assessor variations which may have an impact on the consistency of plantation treatment decisions.

Mechanistic and machine-learning models are available individually, which estimate the environmental disease risk, that is, how conducive the current weather is to disease spread, from temperature, humidity, rainfall and leaf-wetness data, especially for Black Sigatoka [7,8]. In the reviewed literature, however, prediction of environmental risk and severity assessment from images are considered as separate problems, with no identified study incorporating both within a single plant and with repeated time series observations to monitor and forecast the likely change

in severity of an individual plant's disease in the near term. This proposed component explores that particular integration of Cavendish banana.

In the context of this research, field data collection, expert severity validation and acquiring environmental data will be done in cooperation with selected private Cavendish banana plantation. This set includes real Cavendish leaf imagery, agronomic knowledge and field conditions that reflect commercial banana growing, without revealing the name of the estate involved in the collaboration (due to their operation, bananas are only planted on this set if they are in good condition and ripe for harvest).

This research is related to the research cluster of the Centre of Excellence for Artificial Intelligence (CoEAI) and industry vertical from Smart Agriculture / Precision Agriculture. It contributes to Sustainable Development Goal 2 (Zero Hunger) for more timely decisions on disease management, SDG 12 (Responsible Consumption and Production) for the application of agrochemicals targeted by disease severity and SDG 13 (Climate Action) by integrating environmental risk due to climate change into disease decision making.

1.2 Literature Review

We have categorized the literature pertaining to component 3 into six categories: 1) Existing manual severity-assessment standards; 2) Image-based automated estimation of severity; 3) Machine-learning classification of severity to ordinal categories; 4) Deep-learning segmentation architectures; 5) Temporal disease-progression modelling; and 6) Environmental or multimodal methods that integrate image and non-image data. These areas are each discussed thoroughly below and then compared and contrasted in a synthesis table.

1.2.1 Established Manual Severity Scales

The original international scale of banana leaf-spot intensity was proposed by Stover [1] and subsequently revised by Gauhl [2] into a very popular ordinal scale of 0–6 according to the percentage of symptomatic leaf area. Kimunye et al. [3] used this modified scale in a Black Sigatoka distribution survey across Uganda and Tanzania, computing an average of a Disease Severity Index (DSI) per plant for the area, using a formula: $DSI = \frac{\sum(n \times b)}{(N \times T)} \times 100$, where n is the number of leaves within a given grade, b is the grade value (0–6), N is the number of grade categories, and T is the total number of leaves scored; reported field DSI values ranged from about 19.9% to 40.4% by region. For a similar resistance-screening study, Kimunye et al. [4] used the same DSI formulation and also monitored the location of the youngest leaf spotted (as a step progression indicator). Both studies are manually field scored by trained assessors and use no image-based automation, while both demonstrate the exact formula for calculating severity scores that Component 3's Impact score should approximate, which is a computational formula.

Pinzón Núñez et al. [5] developed a modernized and a Cavendish 'Grand Nain' calibrated Standard Area Diagram Set for Black Sigatoka with seven bands of percentage-leaf-area (PLA) (0%, 0–5%, 5–13%, 13–23%, 23–40%, 40–65%, 65–100%) showing better rater precision,

repeatability and accuracy than the older Stover–Gauhl scale, which was reported to have only 71% repeatability in the same comparison. The most compelling evidence that automated, pixel-level estimation of lesion area, in principle, outperforms manual estimation, in that it directly estimates what is being estimated: the percentage of leaf area diseased, as opposed to comparing what is seen to a reference diagram that has been printed.

Chiang et al. [6] studied the Disease Severity Index (DSI) formula that is widely used in all plant pathologies critically, pointing out the statistical problems of treating ordinal severity ratings as continuous data types, and formalizing the standard construction of DSI, which is the sum of the frequency of each class on the DSI multiplied by the score of each class, evaluated and divided by the number of classes. This sets a precedent for the standard use of the weighted-multiplication combination, as well as warning against unexamined arithmetic calculations on ordinal severity grades — a consideration directly relevant to the critical evaluation of the proposed Impact $\times$ Urgency $\times$ Scope formula in Section 3.3.

1.2.2 Image-Based Severity Estimation

To identify Fusarium Wilt and Black Sigatoka affected areas, Elinisa et al. [10] trained a U-Net semantic segmentation model on 18,240 field collected images of banana leaf and stalk, and reported Dice coefficient of 96.45% and Intersection over Union (IoU) of 93.23%. The dataset is specific to bananas, and was collected in Tanzania in the field. The study shows that high accuracy pixel level localization of diseased areas is possible for banana leaves under field conditions but this locates only the damaged parts and does not report a computed lesion-area-to-leaf-area severity percentage; the study also indicates that the disease coverage (Fusarium Wilt, Black Sigatoka) is limited to these diseases while Cordana, BBTv, and Yellow Sigatoka are not covered.

Divyanth et al. [12] developed a two-stage segmentation pipeline that involved first segmenting the leaf versus the background with a U-Net, achieving a mean weighted IoU score of 0.9422, followed by segmenting the leaf lesions within the leaf mask using a DeepLabV3+ model, with the mean weighted IoU score of 0.7379. They computed a ratio of the number of lesion-pixels to the number of leaf-pixels as an indicator of disease severity, which resulted in an R^2 score of 0.96 when compared to the ground truth on a custom dataset of 1,050 corn-leaf images. This is not specific to banana, but the most similar methodological precedent to the proposed banana Scope calculation of Component 3, as it shares the same two-model architecture pairing and quantitatively tests the resulting ratio against expert ground truth.

An open banana-leaf-imagery dataset with 11767 images across three classes (Healthy, Black Sigatoka and Fusarium Wilt Race 1) was published by Mduma and Elinisa [11] and a subset of these images were annotated with LabelMe for performing both mask r-cnn instance segmentation and U-net semantic segmentation experiments. This is a possible candidate source of data for the segmentation model in Component 3, but the underlying pixel-mask files are not publicly available, so it is not possible to verify this data source until the project plan is finalized around it.

An artificial intelligence severity study of *Cercospora* leaf spot in chili peppers [PMC12251717] tested two models, namely Mask R-CNN and YOLOv8, for instance segmentation and classification under field conditions, and determined that Mask R-CNN had higher segmentation accuracy, but underestimated the highest severity (72.3% accuracy in severity III versus 91.4% in YOLOv8), and that YOLOv8 was significantly faster at inference (27ms vs. 89ms). This is pertinent evidence that segmentation accuracy does not necessarily increase with reliability of classification by severity of the data being segmented, and that inference speed is indeed a factor in the trade-off, particularly because the group was required to deploy its Edge-AI solution.

1.2.3 Machine-Learning Severity Classification

A seminal work is by Wang et al. [13] which focuses on severity-category classification using CNN, where they trained multiple CNN architectures to classify apple black-rot images in four severity stages (expert-defined), and the best result, 90.4% accuracy, was obtained by a fine-tuned VGG16 trained with ImageNet. This establishes that transfer-learning based classification for translating continuous disease presentation to a small number of ordinal severity categories is effective directly relevant to the output of Component 3, Low/Moderate/High/Critical.

Kaur (attribution not verified against the publisher record) [14] used classical HSV and Lab colour space thresholding (not deep segmentation) to quantify lesion severity (along with SMOTE-ENN to solve class imbalance issues), and applied four custom CNN architectures for banana disease detection (Cordana, Pestalotiopsis, and Sigatoka), with a CNN of five layers achieving 96.87% accuracy. This is banana-specific and directly relevant but lacks environmental and temporal integration and severity estimation is not based on the lesion area ratio as proposed for Component 3.

The multi-task rice bacterial-leaf-blight framework RCAMNet is used to segment the bacterial leaf blight images into segmentation masks and classify the severity into five stages using a dual-path Convolutional Block Attention Module (CBAM) with a MobileNetV2 backbone. This is not specific to banana, but is applicable as a lightweight, attention-grabbing, edge-deployable classification template that meets the group's requirement to deploy on the edge.

Gradient boosting of trees has been shown to be able to perform tabular severity classification for several crops. A study of severity of a sugarcane foliar disease combined with the measurements of the physiological traits was carried out and the optimized model based on XGBoost had the best performance (precision, recall, F1 and accuracy were all above 0.9186) across four severity levels [PMC12568666]. A tomato late-blight severity study obtained textural and geometric features (contour count, area, perimeter ratios) from the Otsu-thresholded segmented lesions and compared the performance of an XGBoost classifier on five ordinal classes of different severity levels based on the percentage of area affected (0.1 to 10%, 11 to 25%, 26 to 50%, $\geq 51\%$), with Support Vector Machine, Random Forest, k-Nearest Neighbours, and Logistic Regression classifiers using the same data partitioning [AgriEngineering, doi:10.3390/agriengineering8030098]. This tomato study is a fairly close methodological parallel

for the proposed XGBoost stage in Component 3, which also involves segmentation, feature extraction from the segmented lesion, and classification of the lesions to a percentage-based ordinal severity band using the gradient-boosted-tree algorithm.

1.2.5 Temporal Disease Progression Modelling

Pozza et al. [9] simulated Black Sigatoka epidemics at a banana plantation in São Paulo, Brazil for an entire growing season by modelling disease severity (measured by manual scoring) and airborne ascospore dispersal counts over a year long time series, using a monomolecular growth model and finding two peaks of severity. This is banana-specific so it does offer genuine longitudinal severity tracking but not machine learning, and it predicts a population / plantation level seasonal curve, not a per plant short-term change of severity as required for Component 3.

A related study that uses Latent Neural Ordinary Differential Equations [8] predicts the risk of Black Sigatoka infection based on the relative humidity, canopy temperature, and canopy moisture in a continuous-time deep-learning model; however, the authors report that the model was partially trained using synthetic data because there were not enough real datasets with bananas to train the model with. This known limitation in data availability from a study that was trying to forecast banana disease stands as an independent corroboration of the scoping constraint placed on the ambitions of the progression-forecasting of the banana in Component 3.

Comparative time-series forecasting research shows that in small time-series datasets, Gated Recurrent Units (GRU) with fewer parameters and simpler gating structure than Long short-term memory (LSTM) networks are capable of performing at least as well as LSTMs, converge faster during training, and require less memory and computation, whereas LSTM networks have an edge for sequences of large sample size and long horizons with complex long-range dependencies. This comparative evidence will become more relevant as a starting point for the progression-forecasting task in Component 3, though it is empirical and task- and data-specific, and needs to be tested with the project data collected in the field.

1.2.6 Environmental and Multimodal Approaches

A multimodal crop-disease-severity study [18] was motivated by the explicit observation that most previous studies focused on either images or environmental data to predict crop type, presence and severity of disease, using a data-augmentation strategy called mixup to combine image and environmental information. However, while it may not be specific to banana, it is a direct precedent for the integration of image and environmental modalities and the mixup augmentation technique is transferable as in Component 3 the training set is likely to be small and highly imbalanced multimodal. The combination of diagnosis and severity estimation has also been seen in another comparable study of tomato [19] that utilizes LSTM networks to predict the severity of the disease, specifically by combining weather time-series data, which also sets precedent for the combination of temporal data and environmental data proposed here.

A recent systematic review of plant-disease forecasting models [20] shows that recurrent architectures are generally used to model the temporal dynamics of an epidemic based on weather and historic disease data, that simpler threshold-accumulation models are operationally competitive when disease data is sparse, and explicitly recognizes the need for longer series of high-frequency, standardized disease observations as a current limitation across the field. This independently verifies the Data-availability constraint expected for the temporal and forecasting sub-tasks of Component 3 and supports a conservative initial scoping of the forecasting approach, as described in Section 3.4.

In addition to the preceding general example of risk assessment in plant pathology, there is a general example in other fields of science, as well. Pawar and Shinde [24] developed a wireless-sensor based decision support system for powdery mildew monitoring which calculates a Risk Index based on several weighted environmental data such as temperature, soil moisture, relative humidity, wind speed and light intensity. This is not banana specific at all, and is not directly formalized by the Impact $\times$ Urgency $\times$ Scope formula itself, but it shows that the multi factor weighted combination for agricultural risk/severity scoring is a well-established general pattern.

Table 1 — Overall Existing Research Comparison Summary

Reference	Severity Scoring	Image-Based	Environment	Temporal Tracking	Forecasting
[3],[4]	Yes (manual DSI)	No	No	Partial	No
[5]	Yes (manual SADs)	No	No	No	No
[10]	Partial (localization)	Yes	No	No	No
[12]	Yes (ratio-based)	Yes	No	No	No
[13],[14]	Yes (categories)	Yes	No	No	No
[7],[8]	No (risk, not severity)	No	Yes	Partial	Partial (risk only)
[9]	Yes (manual, seasonal)	No	Partial	Yes	Partial (seasonal curve)
[18],[19]	Yes	Yes	Yes	No / limited	No
Proposed Component 3	Yes (automated)	Yes	Yes	Yes	Yes (short-term, conditional)

1.3 Research Gap

There is a good track record of research in a number of discrete areas relevant to Component 3. A scale for Black and Yellow Sigatoka exists which is both established and quantitative, the Stover–

Gauhl scale [3],[4],[5]. Banana disease localization has been shown to be feasible by automated image-based segmentation of lesion areas [10],[11] and for two-stage computation of the leaf/lesion severity ratio in other crops [12]. For Sigatoka diseases there is existing knowledge of environmental infection-risk modelling [7],[8]. severity classification using machine learning methods based on raw images and tabular features extracted from images has been proven in various crops [13],[14],[PMC12568666],[AgriEngineering]. For other crops, multimodal fusion of image and environmental severity has been shown [18],[19].

There are a few caveats, though. Usually, disease identification and severity assessment are separated; most of the research about bananas and AI is about determining the type of disease, but not the severity of a known disease. A few image-only severity approaches have not included environmental aspects and environmental-risk approaches have not quantified the severity of what is currently seen in the image; the two types of research have largely been independent. In contrast to the two Sigatoka diseases, no validated, quantitative standard of severity could be found for Cordana or BBTv. Where available, temporal progression modelling is conducted in plantation or seasonal time frame, not at short plant-specific intervals, and even a dedicated banana infection-risk forecasting study had to use synthetic data because of the lack of real data. During this review, public banana datasets were found to include disease classification labels, but not severity labels, masks of the lesions at the pixel level nor repeated observations over time, which means that these data cannot be used directly in severity or progression modelling without additional expert labeling.

No identified study, therefore, integrates image-derived lesion area severity, environmental risk and repeated temporal per plant observations into an evaluation framework for banana disease; there is also no identified study of banana disease that forecasts disease progression status for individual plants at a temporal scale small enough to support individual plant treatment prioritization. This is important because a per-plant, “near real-time” indicator of the current severity and likely future course of the disease, as provided by Component 4 in the overall system, is needed to support the prioritization, which cannot be addressed directly by a system that could only classify the disease or provide a seasonal risk at the population level.

Research Gap Statement: Limited integration of images-based lesion area severity, environmental risk information and temporal observation history within a single framework for Cavendish banana disease assessment and no quantitative severity standard exists for Cordana and BBTv specifically. It looks into this integration and suggests practical approaches to assessing the severity of the two diseases without a validated standard in component 3.

2. OBJECTIVES

2.1 Main Objective

To develop and test a multimodal disease severity assessment and short term forecasted progression assessment module for Cavendish banana incorporating image-based lesion area severity, environmental risk information provided by Component 2, and an observation history to generate a quantitative severity score, a validated agricultural-expert based assessment of

Cavendish banana disease severity was collected in collaboration with a selected private Cavendish banana plantation to support the development of the image based assessment.

2.2 Specific Objectives

To develop and test a pipeline for segmentation of leaves and lesions from Cavendish banana leaf images based on the U-Net and calculate the lesion-area-to-leaf-area percentage (Scope) for leaf images, in comparison with similar published results for leaf/lesion segmentation using Intersection over Union and Dice coefficient. The aim of Objective 2 (Disease-Specific Impact Scoring) was to implement and validate an automated Impact score for Black and Yellow Sigatoka, using the previously established percentage-band structure of the disease developed by Stover and Gauhl (Stover–Gauhl/SADs), and to propose and pilot practical proxies for severity assessment for Cordana and BBTv, for which a quantitative standard does not exist.

Objective 3 — Severity Classification and Regression: Design and train an XGBoost severity model which outputs a continuous severity score and classifies this score to Low-Moderate-High-Critical classes based on image-derived, environmental and metadata features, with both regression and classification evaluation metrics.

Objective 4 — Environmental Integration and Fusion Evaluation: Define an input contract for consuming priority scores from the environmental component (Component 2); Critically analyse the proposed Impact $\times$ Urgency $\times$ Scope fusion formula and alternative fusion strategies based on expert-assigned priority scores as ground truth.

Objective 5 — Temporal Progression Tracking and Forecasting: To build a progression-trend classifier between Worsening, Stable, and Improving states of repeated plant observations, and to evaluate the possibility of short-term severity forecasting using a GRU model, compared to a rule-based and regression baseline, where anticipated data-volume constraints are anticipated.

Objective 6 — Expert Validation: To validate model generated severity scores, categories, and progression trends with agricultural-expert scores and categories derived from the collaborating private Cavendish banana plantation using rank correlation and standard classification and regression metrics.

3. METHODOLOGY

3.1 Research Methodology/ Design

Research questions in Component 3 are largely quantitative and predictive: Can lesion-area measurements derived from images be reliably mapped onto severity measurements defined by other conventional or proposed severity scales? Can the use of image, environmental and temporal information improve severity assessment over any one of these information sources? Can short-term severity progression be predicted, based on limited repeated observations? These questions call for quantitative, measurable, comparable outputs such as segmentation accuracy, classification and regression performance, and forecast error, not for an interpretive or exploratory, qualitative design. In this case a quantitative, experimental research methodology is used, with a supervised deep learning algorithm for segmentation, a supervised machine learning algorithm for

classification and regression, and a time-series model for progression forecasting, which is assessed by controlled comparison with baselines and experts' judgment. This methodology is supported by four arguments. First, the character of the research question, which is one of severity and progression, is a numerical or ordinal quantity that must be predicted numerically or otherwise, not described qualitatively. Second, the data type available: image, tabular environmental, and repeated-observation time series data all have existing quantitative methods that are appropriate. Secondly, the nature of the data: Image data, tabular environmental data, and repeated observation time series data, respectively, are well suited to established quantitative methods. Thirdly, the experimental requirements: model and fusion strategy selection can not be justified without a controlled comparison with baseline models and fusion strategies, where image-only, environment-only, and rule-based fusion approaches are compared. Fourth, all the supporting literature adopts quantitative/experimental approaches to similar problems of estimating severity and predicting progression (as in [10],[12],[13],[18],[20]).

3.2 Data Sources and Data Requirements

The following four categories of data are needed for the Component 3: leaf images with the classification of disease, disease severity labels, pixel level segmentation annotations, environmental risk information, and repeated temporal observations of the same plants. Each is described below, along with the amount of the material that is available from public sources and that needs to be collected specifically for experts to annotate.

3.2.1 Leaf Images

The leaf images will be acquired from the collaborating private Cavendish banana plantation through mobile-device photography under field conditions, with the help of public banana disease datasets for pretraining and augmentation if necessary. The images will include all four target diseases (Black Sigatoka, Yellow Sigatoka, Cordana, BBTv) and healthy controls, but will be focussed to include a range of severities rather than only late stage severities, as the ability to determine the severity is of greater practical value for early intervention. The image quality should include sufficient resolution to see individual lesions, consistent framing of the leaf surface for later evaluation, and capture conditions (lighting, time of day, camera device) for future evaluation of field-robustness.

3.2.2 Disease Labels

The detection output of images processed through the integrated system will be read by disease classification labels that were produced in Component 1, while for images collected independently from Component 1, labels will be confirmed by agricultural experts at the collaborating plantation, during the training phase. Disease label and severity label are different — A disease label indicates what disease is present, a severity label indicates how far the disease is in the severity stage, public datasets that were confirmed to contain the former were not confirmed to contain the latter (Section 3.2.7).

3.2.3 Severity Labels

For the supervised learning of the severity classification and regression models, the severity data is needed that are labelled with severity grades, percentages or categories per image by an expert. These labels will be created via expert annotation by applying the Stover–Gauhl/SADs percentage bands for Black and Yellow Sigatoka and the proposed proxy scales (Section 3.3) for Cordana and BBTV. Because of the practicalities of a one-supervisor-team annotation process, inter-expert agreement will be discussed in cases of multiple expert annotators, but not in cases of a single expert, in which case the lack of inter-expert agreement will be stated explicitly—rather than being implied by the discussion of an inter-expert agreement statistic.

3.2.4 Segmentation Annotations

To train the U-Net segmentation model, masks of the leaf area and leaf masks with the lesions are needed for a representative sample of images collected—leaf masks for delineating diseased tissue from healthy leaf tissue and pixel-level leaf masks for delineating leaf tissue from background. These annotations will be created with the help of CVAT or Roboflow by tracing leaf and lesion outlines, with expert input as to which parts of the leaf or lesion are part of the disease and which of incidental damage or senescence where possible.

3.2.5 Environmental Data

Component 2 gathers and analyses environmental data pertinent to banana disease risk, namely temperature, relative humidity, rainfall, soil moisture, and leaf wetness, to generate a zone level environmental risk or urgency score. Component 3 does not gather or model environmental data, but simply takes the output of Component 2 as the Urgency data fed into the severity scoring process outlined in Section 3.3, organised by plantation zone and observation date.

3.2.6 Temporal Data

To support progression-trend classification and forecasting, it is needed to observe the same plant over time. Ideally each observation record will include an image, a derived severity score, the environmental risk score, the date of the observation and a persistent plant or leaf identifier. The plant or leaf identifier is necessary because only repeated measures of the same physical entity can be used to determine progression, and without the identifier, repeated images could not be differentiated from independent single time-point images, and modelling progression would be impossible.

3.2.7 Data Combination Strategy

No single data source contains all the information needed so a combination strategy is used to connect records between data sources by using a common plant identifier, plantation zone, and observation date or timestamp. Public image datasets provide information on the disease/imagery which can be used for pretraining and augmentation. The collected private-plantation subset is annotated with severity labels and segmentation masks by experts. Images from the collaborating plantation are captured in the field to provide real world images, and if a re-visit of the same plant is conducted, the repeated observation is captured. The Urgency component of the severity framework comes from environmental data received from Component 2. These are linked

together by creating a multimodal data structure like: Plant_001 + Date_01 + Image + Severity + Environmental Risk and then Plant_001 + Date_02 + Image + Severity + Environmental Risk, and so on for successive observations. A series of dated records with a particular plant identifier creates a strength of evidence for modelling the temporal progression, as described in Section 3.4.

When you are combining the sources, you have to take into account a number of issues that might not be assumed obvious. The resolution and capture device of the images in public databases and plantation-collected images are different, so there is a need for standardisation before preprocessing. There are a number of differences in the disease-label vocabulary across the public sources, such as the 'Sigatoka' label appearing as one in some sources but not the other, which will need to be carefully mapped to the four-disease taxonomy used by the Component 1. There is some vocabulary that is ambiguous, and this must be carefully mapped with lower-confidence mappings explicitly documented. Geographic origin and banana cultivar variation within public datasets, most of which were collected outside Sri Lanka, and cultivar variation between the public datasets and the Cavendish cultivar cultivated at the collaborating plantation pose a domain-shift risk that requires the pretrained models to be fine-tuned for, but not assumed to transfer naturally. The plantation collected, expert-annotated dataset will then be used for primary training and evaluation of the severity model, temporal model and fusion model, while the public data will be used for pretraining and augmentation of the segmentation and classification backbones.

3.2.9 Data Preprocessing

Table 2 — Data Preprocessing Pipeline Steps

Step	Purpose
1. Data collection	Acquire leaf images and repeated observations from the collaborating plantation and public sources
2. Data cleaning	Identify and correct corrupted files, mislabeled samples flagged by experts, and format inconsistencies
3. Image quality checking	Filter blurry, poorly lit, or heavily occluded images that would corrupt lesion-area computation
4. Duplicate removal	Detect and remove duplicate or near-duplicate images, particularly where public datasets share source material
5. Image resizing	Standardize images to a consistent resolution for model input
6. Normalization	Normalize pixel intensity values and correct for lighting variation across capture conditions
7. Data augmentation	Apply rotation, flipping, and colour-jitter augmentation to increase effective training-set diversity, particularly for under-represented disease classes

8. Annotation checking	Verify segmentation mask and severity-label consistency, flagging outliers for expert review
9. Segmentation mask preparation	Convert polygon annotations into pixel-level binary masks suitable for U-Net training
10. Feature extraction	Derive lesion-area percentage, disease-specific Impact score, and other tabular features from segmentation output
11. Environmental data alignment	Match Component 2's environmental risk score to each image record by zone and date
12. Missing-value handling	Apply documented, justified imputation or exclusion rules for missing environmental readings or incomplete records
13. Temporal alignment	Order and link repeated observations per plant identifier into chronological sequences
14. Dataset splitting	Split data at the plant level into training, validation, and test sets to prevent data leakage (Section 3.8)

3.3 Severity Calculation Framework

The proposed severity framework is built around three components – described individually below – of impact (disease severity as defined in plant pathology); scope (proportion of leaf area affected as determined by image analysis); and urgency (environmental disease risk received from Component 2). Each component is described in detail and the suggested approach for fusion is explained along with its limitations.

3.3.1 Impact

For Black Sigatoka and Yellow Sigatoka, the Impact will be calculated as per the original formulation from the Stover–Gauhl Disease Severity Index [5] where the leaves are assigned an ordinal disease rating from 0 to 6 depending on the proportion of leaf area with symptoms and the grades are added together and multiplied by the appropriate bands to give the DSI: $DSI = \frac{\sum(n \times b)}{(N \times T)} \times 100$, where n is the number of leaves in a given grade, b is the band, N is the total number of leaves in the set, and T is the number of bands. In the case of an automated implementation, the lesion-area percentage resulting from the segmentation (Section 3.3.2) will be mapped onto the same percentage bands, without calculating an independent severity estimate, in order to be consistent with the established and expert-validated standard. For BBTv, which has no published quantitative severity band structure, a symptom-stage ordinal classification (vein streaking, chlorosis, leaf shortening, stunt and bunching) is proposed, as BBTv does not have a localized-lesion biology like Cordana; for BBTv that also had no lesion-area percentage band structure, a disease-specific proxy is proposed: leaf shortening, chlorosis, vein streaking, bunching and stunt.

3.3.2 Scope

The scope is computed as the percentage of the leaf surface with visible lesions which can be directly obtained from the image segmentation: $\text{Scope (in \%)} = (\text{Lesion Area} / \text{Total Leaf Area}) \times 100$. The U-Net segmentation model generates two separate masks at the pixel level for a given image: a leaf mask which segments the leaf area from background and a lesion mask which segments the diseased area within the leaf area. The total number of leaf-area pixels and the number of lesion-area pixels is summed from the respective mask and the ratio (as a percentage) of the two sums is the Scope value. This calculation is similar to that performed by Divyanth et al. [12] for corn disease severity, where the same ratio was shown to be consistent with human experts ($R^2 = 0.96$).

3.3.3 Urgency

The output of Component 3 is not the computation of urgency, but the input from Component 2 environmental disease-risk prediction for the zone of the plantation and the date of observation. Conceptually different to Impact and Scope, Urgency refers to the degree to which the environment is conducive to further disease development, at present or recently. A plant may have a high severity now when the environment is not so favourable, or low severity now when the environment is quite favourable and it will spread quickly in the future; Urgency is not a repeat of the Impact and Scope severity measurement, but has been added to capture the forward looking dimension.

3.4 Proposed Fusion Formula and Its Limitations

The severity framework being investigated looks at the three components with the formula: $\text{Final Severity Score} = (\text{Impact} \times \text{Urgency} \times \text{Scope}) / 10,000$; each one of the three components are scaled from 0 to 100. Of the many plant-pathology and agricultural studies that were reviewed, no single study included an image-derived Impact and Scope score multiplied by an independent environmental Urgency score, in this particular multiplicative format. The combination of the three into a single multiplicative score is the proposed multimodal fusion framework that is under investigation in this research, as opposed to being taken directly from an equivalent published precedent, and is grounded in the well-established Stover–Gauhl/SADs standard (Section 3.3.1) and in the well-established image-based affected-area measurement principles demonstrated in prior segmentation-ratio research (Section 1.2.2), all of which are used for environmental risk assessment within the scope of agriculture and other disciplines. A multiplicative structure is known to have a particular limitation, documented in the multiplicative risk-scoring literature: If one factor is approaching zero, the composite score is approaching zero even if the other factors are at their peaks. If applied to this situation, a plant with high Impact (high visible lesions) and high Scope (high environmental risk) and currently rated low Urgency (low environmental risk) would be given a low combined score even when it is rated as having severe lesions. It's not a minor point, just a methodological limitation, as this is a combination of the severity of the plant at this moment and the probability of aggravation, which are two different things that are relevant to different decisions. Further, the impact and scope as defined, for lesion based diseases are not entirely independent of each other as Scope is a direct measurement of percentage-affected-leaf-

area which is also the basis for impact, while impact is derived in the form of percentage-affected-leaf-area bands. Alternative fusion strategies to be explored via experiment are: weighted-sum combination that adds the three factors (Impact, Urgency, Scope) instead of multiplying them with learned or expert-validated weights, avoiding the zero-collapse limitation; a regression-based fusion model that learns to predict expert-assigned overall severity or priority rankings directly from the three inputs; and a lightweight machine-learning fusion model, if sufficient paired data is available. The current severity (derived from Impact and Scope, which describe the plant's present condition) and environmental Urgency are to be reported separately, clearly labelled, and combined multiplicatively only at a final prioritization stage for the resource ranking function in component 4 with a particular forward-looking value of multiplication, suitable for a severity score that describes present condition, not the potentially amplifying value of multiplication. This final fusion approach will not be taken for granted, but will be chosen depending upon experimental results based on expert-determined severity and priority rankings, measured through Spearman's rank correlation.

3.5 Severity Category

The continuous severity score will be mapped to an interpretable scale (Low, Moderate, High, Critical) for use in the field as well as being part of the numerical score used internally for ranking by Component 4. In the case of the computed severity score of 72, it may be in the High category. The score gives detailed information that is precise and can be used to make comparisons that can also be used for trend analysis and ranking; this is useful for numerical interpretation by the users of the fields. The score gives an immediately interpretable summary without requiring any numerical interpretation by field users. Initial thresholds of categories are proposed as a provisional guide, based on the Horsfall–Barratt scale and the European and Mediterranean Plant Protection Organisation scale PP 1/181 that recognise 5% and 15% affected-leaf-area as thresholds between low, moderate and high economic-risk categories for foliar diseases [25]. These provisional thresholds should not be used as final thresholds without expert validation, a review of the available literature for each of the four target diseases, and a statistical analysis of the distribution of severity scores collected.

3.6 Temporal Progression

The progression is determined by comparing a plant's score with a previously recorded score or scores for the same plant identifier. A Worsening classification is assigned if severity scores increase successively over time, e.g., if a plant had a severity score of 25% at Day 1, 38% at Day 7, and 52% at Day 14. If a validated tolerance limit is defined for the stable and the measurement noise, a comparatively stable trend, such as 48% at Day 1, 49% at Day 7 and 48% at Day 14, can be classified as Stable. A negative trend overall would suggest that there is a trend of Improvement. There are no a priori assumption of the numerical thresholds separating the categories Worsening, Stable and Improving, but the thresholds have to be based on the statistical distribution of the severity score changes observed in the temporal data set collected, and be checked against expert judges' considerations of how much the severity score changes means in terms of the field situation. Secondary objective of short term forecasting is to predict the severity category of a plant for a time period of around 7 days ahead, given the severity sequence which

has been experienced by the plant in the past and, if available, its recent environmental risk history. Based on the expected limited number of repeated observations to be made over a single field season, the forecasting will be carried out using a rule-based extrapolation of the observed trend and a simple regression baseline, and only if a large number of sequence observations is accumulated during a field season will the GRU-based model be explored, following the evidence of data scarcity presented in Section 1.2.5.

3.7 Model Architecture

3.7.1 U-Net Segmentation

Image segmentation is the classification of an image into segments at a pixel level representing different semantic categories of pixels, and different from image classification, which assigns a single label to an entire image and different from object detection, which localises objects using bounding boxes, but not at a pixel level. The specific category relevant here is semantic segmentation, where each pixel in an image is assigned to one of the classes (e.g. leaf tissue, lesion tissue, background), and is required for Component 3 as the calculation of Scope (Section 3.3.2) requires a precise number of lesion tissue pixels compared to the total number of leaf tissue pixels, which neither classification nor bounding-box detection can directly give.

U-Net is a symmetric convolutional neural network (CNN) architecture, composed of an encoder and a decoder, with skip connections, that was developed originally for biomedical image segmentation. The encoder path slowly reduces the spatial resolution of the input image, while gaining more and more abstract feature representations by using several convolutional and pooling layers. These features are upsampled progressively along the decoder path to the resolution of the original image, which results in a pixel-level segmentation output. Skip connections skip through the same level of the encoder and decoder, connect at the same resolution, and let the fine-grained spatial detail lost during downsampling be recovered during upsampling, which is crucial for accurately identifying the boundaries of irregular regions of fine-grained lesions as opposed to coarse leaf outlines.

The literature review in Section 1.2.4 has shown that U-Net has proven to be very successful in directly segmenting banana leaf images with high accuracy [10] and that U-Net was superior to DeepLabv3+ and SegFormer in segmenting diseased regions near the leaf edge and tips under the conditions of plantation field photography with complex background. In addition to standard preprocessing like resizing, normalization and augmentation (Section 3.2.9), paired leaf and lesion segmentation masks are required for training, which are generated by expert-guided annotation (Section 3.2.4). The model provides the predicted masks for leaves and lesions from a given image, which are then fed into the Scope calculation. The quality of the segmentation is assessed by Intersection over Union and Dice coefficient, which are explained in further detail in Section 3.9. The selection of U-Net as the first architecture is not considered as a final one and is compared with alternative architectures in Table 4, when plantation data is available for real applications it is further evaluated and then considered as one of the final architectures.

3.7.2 XGBoost Severity Prediction

XGBoost (Extreme Gradient Boosting) is a decision tree-based ensemble learning algorithm that constructs a series of shallow decision trees, each of which learns the model's errors from the previous ones, with gradient boosting optimization and regularization to prevent overfitting. XGBoost is suitable for tabular data, like structured data, but not for raw image data; this is why it is proposed for the severity-scoring stage, which works on features from the segmentation and environment stages, not on raw image data directly.

The lesion-area percentage (Scope) from segmentation is proposed as a possible feature for input, along with the disease specific Impact score, the environmental Urgency score from Component 2, the detected disease class, and, if applicable, other image-derived features such as average lesion size or lesion count, as exemplified in the tomato late blight severity study reviewed in Section 1.2.3. To ensure consistency between the regression and classification output, as discussed in Section 3.5, the model will be trained to perform two related tasks: a regression task, which predicts the severity score (a continuous value), and a classification task which predicts a discrete severity category (a hard binary classification), the latter of which is obtained by thresholding the former.

On the basis of comparative evidence provided in Section 1.2.3, where XGBoost consistently outperformed or performed as well as other classifiers including Random Forest, Support Vector Machine, k-Nearest Neighbours and Logistic Regression when applied to a variety of crop severity studies using similar tabular, feature-derived input data, XGBoost is proposed as the initial modelling choice. XGBoost also offers explanations of feature importance, a feature that will enable agriculture end-users to understand the severity prediction of the system—conforming to the overall system's stated focus on explainable AI. Random Forest and Support Vector Machine will not be eliminated from the pool of models but will be considered as comparison baselines in model selection.

3.7.3 GRU Progression Forecasting

Sequential data are those that are measured in such a way that the order of the observations provides some useful information, e.g., a history of severity for a given plant provides information regarding the likely progression of severity values not provided by an unordered set of the same values. To solve this, a recurrent neural network (RNN) is suggested over the classical feed-forward model to model the progression of the disease based on a sequence of observations, rather than to treat each observation independently.

A variant of a recurrent neural network which keeps a hidden state updated at every sequence step, with the use of an update and reset gate to determine whether to retain most of the previous hidden state or to include more of the previous input information—without the separate memory cell state of the related Long Short-Term Memory architecture. This simpler gating mechanism results in fewer trainable parameters of GRU than those of an equivalently sized LSTM, and in at least several comparative studies, the LSTM was found to achieve generalization at least as well in small datasets, or to train faster, but the ability of LSTM to learn faster on long, more complex sequences, with significant long-range dependencies, is its main advantage.

In the case of Component 3, the input from the GRU is a time series of previous observations of severity, and the expected output is the forecasted value of the severity at a short time horizon, around 7 days ahead. GRU is proposed as the architecture for investigation upon the accumulation of a sufficient amount of temporal sequence data, not as the first architecture to be implemented; simple regression baseline and rule based trend extrapolation are proposed as the initial forecasting approach with the addition of GRU if it turns out to offer a measurable improvement on the actual data collected. Alternatives to the two sequence architectures mentioned above are still available, such as the Long Short-Term Memory and Transformer sequence architectures, with the latter being cited in the literature as requiring “significantly larger” training sequences than GRU or LSTM, which is not compatible with this project's intended amount of data.

3.7.4 Multimodal Fusion

Multimodal fusion involves the fusion of information from different data modalities, such as image-derived features, environmental risk, and temporal sequence information, to generate a single prediction. There are three main general types of fusion found in the literature: early fusion in which raw or lightly processed features from each modality are concatenated prior to passing them to a single model; late fusion in which features from each modality are linearly processed by separate models and then merged in the output; and intermediate fusion which fuses partially processed feature representations from each modality within an intermediate layer of a joint model [17].

With the expected size of the data set to be collected at the plantation, early and intermediate fusion are expected to require large paired training sets across all modalities, which is not likely to be available. Late, decision-level fusion is therefore proposed as the first approach: The output of the image-processing stage (Impact and Scope) is computed largely independently, while the output from the Component 2 (Urgency) is computed, and the output of the temporal-modelling stage (trend) is combined at a final stage using the fusion approach selected in Section 3.4. This is in line with the overall result that has been found in the Plant Pathology review of image-based crop-disease detection [17] that late fusion is more feasible when there is limited multimodal data. The inputs for the fusion stage are then the Impact, Scope, Urgency, and progression-trend values for a particular plant observation, and the output of the fusion stage is the final severity score, severity category, and progression indicator passed on to Component 4.

3.8 Model Training Process

The input to training of the U-Net segmentation model consists of paired leaf images and their respective expert-annotated leaf and lesion masks, which are then fed into and trained with a segmentation loss function (e.g., a combined Dice and cross-entropy loss) to minimize the difference between the ground truth and the predictions for the segmentation mask, and its output is a predicted segmentation mask for a new image, from which the percentage of lesion area is calculated.

The training data for the XGBoost severity model includes expert-labeled feature records, which consist of the lesion percentage, Impact score for the disease, Urgency score for the environment

and image-derived feature descriptors that are relevant to the disease. The model is trained to predict a continuous severity score; the expert-assigned severity category or percentage is used as the prediction target. The output of the model is a predicted continuous severity score, which can be transformed into a predicted severity category by thresholding.

The GRU progression-forecasting model is trained on a sequence of observations of the same plot or plant with all observations listed sequentially in terms of severity and optionally environmental, values across different dates, and the forecast target is a future severity value or category for the plant over a defined short-term forecast horizon; its output being a probability distribution over the plant's likely state of severity at the defined short-term forecast horizon.

For all three models, data will be split into training and validation data at the plant level and test data, as explained in Section 3.9, so as to avoid the same plant appearing in the training and validation sets. The primary train/validation/test split will be complemented by "grouped" k-fold cross validation (where applicable based on the size of the dataset) by plant identifier. The model will only be tuned using the validation set and test set will be used only for final unbiased reporting of the performance of the model. Standard techniques suitable for each model family will be employed to mitigate overfitting: dropout and early stopping for U-Net and GRU models, and bounded tree depth and learning rate, plus regularization for the XGBoost model.

3.9 Baseline / Benchmark

The four categories of baseline will be used as the benchmark for Component 3. The manual scoring system from field experts by the man, the Stover-Gauhl/SADs, is, of course, the current practice, and the standard that the automated system is meant to reach or surpass in consistency, if not speed. Image-only severity estimation is performed without any environmental or temporal input, and it tests whether or not visual information alone is sufficient. Environment-only severity proxies test whether the risk score from Component 2 alone is a reasonable approximation of the visible severity in the context of the environment. A minimal complexity comparator is a simple rule-based severity scoring where a disease-specific weight is assigned for the presence of the disease and is not segmented. A multimodal fusion approach must perform better than a multimodal baseline on the metrics outlined in Section 3.10, which explicitly requires comparisons to a baseline, to support the added complexity of the multimodal integration.

3.10 Evaluation Metrics

Table 3 — Evaluation Metrics

Metric	Why Relevant	Target / Comparison
IoU / Dice (segmentation)	Measures pixel-level accuracy of leaf and lesion masks, the basis for the Scope calculation	Benchmarked against comparable published leaf/lesion segmentation results
Precision / Recall (segmentation)	Measures over- and under-segmentation of lesion regions specifically	Balanced precision and recall on held-out plant-level test set

MAE / RMSE / R ² (severity regression)	Measures continuous severity-score prediction accuracy against expert-assigned ground truth	Outperform image-only and rule-based baselines
Accuracy / Precision / Recall / F1 / Macro-F1 (severity classification)	Measures Low/Moderate/High/Critical classification quality; macro-F1 accounts for expected class imbalance	Outperform baseline classifiers; macro-F1 prioritized over raw accuracy
Confusion Matrix	Identifies which severity categories are most frequently confused	Used diagnostically to guide threshold refinement (Section 3.5)
Trend classification accuracy	Measures Worsening/Stable/Improving classification quality	Outperform majority-class baseline
Spearman's rank correlation (expert validation)	Measures agreement between model-ranked and expert-ranked plant priority	Statistically significant positive correlation with expert rankings

3.11 Validation Strategy

Plant-level, or subject-level, train, validation, test splitting will be used to ensure that all images for a particular plant or leaf identifier are in one split, which prevents the model from being trained on the specific identifiers' visual characteristics (background, lighting, leaf shape, etc.) rather than the severity information. When evaluating the model for progression-forecasting, a forward-chaining temporal split will be adopted, where timepoints earlier in the observational period will be used for model training and timepoints later in the period will be used for model evaluation, because otherwise a model could be evaluated with data that it would not have had available during actual operation. A train / test split will be used as the primary train / test split, with k-fold cross validation (in groups by plant identifier) included where the size of the dataset allows. Model outputs will also be compared directly with expert severity estimates, and, where possible, to expert overall plant-priority rankings, acquired via the partner private Cavendish plantation, through Spearman's rank correlation. This adds the requirement for expert validation as part of the existing task definition in Component 3, and offers an intelligible, non-technical result of this validation which can be communicated to agricultural officers not experienced with machine-learning specific evaluation metrics. Expert-model disagreements will not be ignored or dropped, but will be recorded and considered; systematic disagreement patterns (e.g., all experts consistently underestimate the severity of models for a particular range of severity) may be indicators of revising the severity-mapping bands described in Section 3.3.1, rather than random evaluation error.

3.12 Development Approach

Component 3 will use an iterative, Agile-informed development approach with a focus on the evolving inputs derived from Component 1's evolving disease classification, and Component 2's environmental risk score, both of which will be in concurrent development by other members of the research group. Development will be done in short cycles: beginning with a classical approach to segmentation and non-deep approach to rule-based severity baseline development and then proceeding to deep segmentation development and forecasting models with GRUs,

which would require much more data to be annotated and would only be worth it once everything else was proven possible. This is the standard approach to testing a basic model first and increasing the model complexity over time; it recognizes that the amount of annotated data will be collected in the field in an incremental way rather than all in one time. The proposed solution will provide an overview of the components involved

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Components

The overall group solution is a 4 component framework to support decision making for Cavendish banana disease management that is deployable on Edge-AI. In Component 1, early disease detection is achieved from leaf images using few-shot learning and explainable AI, providing a label of the disease class and localization of infected region. Component 2 forecasts the environmental disease risk based on temperature, humidity, rainfall, soil-moisture, and leaf-wetness data from the sensors, resulting in a per-zone risk or urgency score. Both of these outputs are used in the third component (this proposal) which also includes re-sampled observations of the same plants over time, and yields a quantitative index of severity, an understandable category of severity and a trend of plant progression, for which short-term forecasting is explored when possible because of the data. Component 4 takes outputs from Component 3 to prioritise plantation zones and to suggest resource allocation like pesticide use, labour allocation and scheduling of inspections.

Overall System Overview

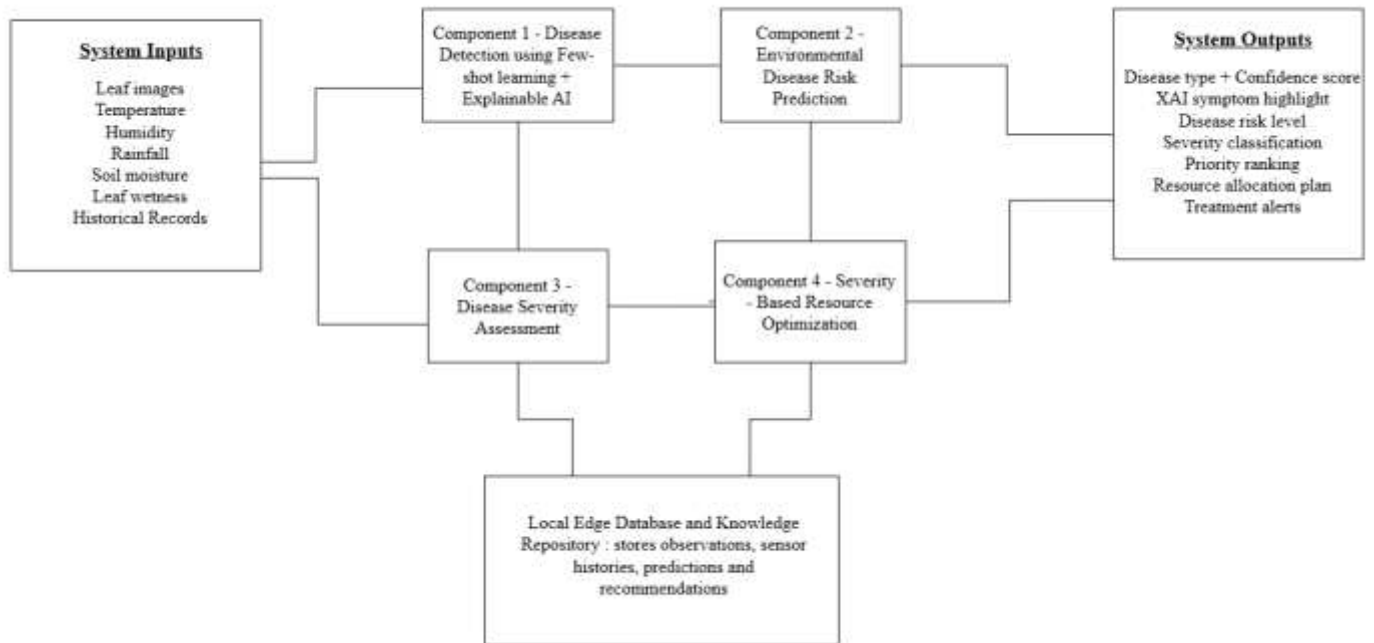


Figure 1 Overall System Overview

Individual Component System Diagram

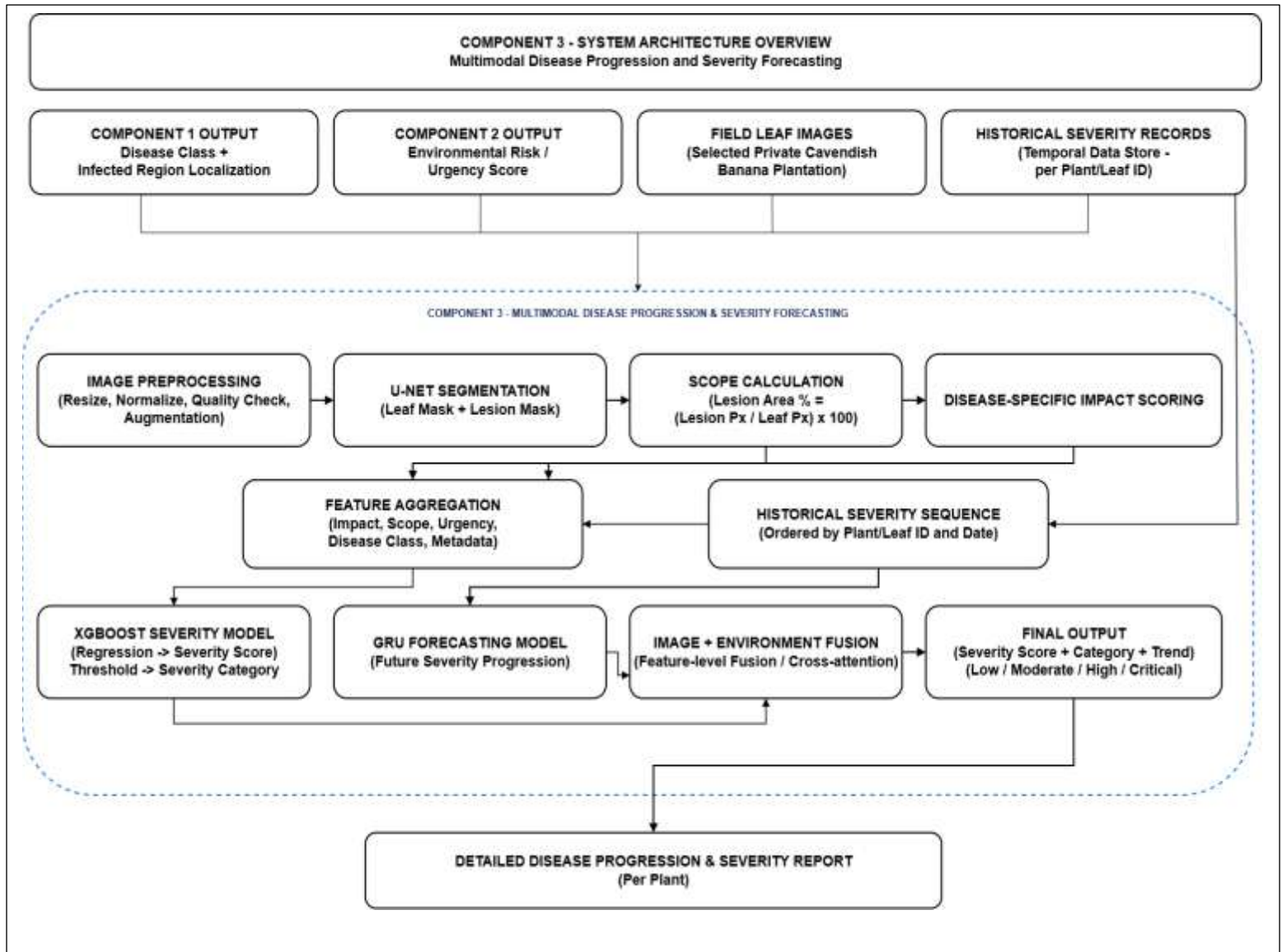


Figure 2 Component System Diagram

4.2 Overall Integration

Data and knowledge are shared between the four components but not transmitted from component to component. Component 3 takes in the disease classification from Component 1 and environmental risk score from Component 2 for the specific plant or zone and date, and outputs the disease severity score, disease category, lesion-area-percentage, and progression-trend to be used by Component 4. Due to the group's deployment need for Edge-AI, this loosely coupled repository-driven design enables each component to be independently developed and tested.

4.3 Dependencies Among Components

Component 3 will rely on the output format of Component 1, and will require localization of infected regions for seed-based lesion segmentation – a contract to be agreed with the member responsible for Component 1, before the input-processing stage of Component 3 is completed – and the environmental Risk Score output format of Component 2 with numeric range, spatial granularity and temporal update frequency to be agreed with the member responsible for Component 2, and the shared local database or repository structure and access method to be agreed by all four participants.

4.4 Planned Integration Method

The approach for integration will be staged: one component will be developed and tested with synthetic or held-out sample inputs that adhere to the agreed contract format, followed by integration testing two-by-two before full four-component integration testing with the shared Edge-AI deployment target. Version control will be maintained in a shared repository and interfaces between components will be fixed as an input and output schema to minimise the risk of integration, in view of the parallel development time line of the group.

4.5 Individual Responsibility Boundary

Table 4 — Individual Responsibility Boundary

Activity / Deliverable	Individual Responsibility	Shared Responsibility	Other Member
U-Net leaf/lesion segmentation model	Yes		
Lesion-area (Scope) calculation	Yes		
Stover-Gauhl/SADs-based Impact scoring	Yes		
Cordana/BBTV severity proxy design	Yes		
XGBoost severity classification/regression	Yes		
GRU progression forecasting	Yes		
Impact-Urgency-Scope fusion evaluation	Yes		
Expert validation of severity outputs	Yes		
Disease classification / detection			Component 1 member
Environmental risk score generation			Component 2 member

Resource allocation / prioritization logic			Component 4 member
Shared database/repository structure		Yes (all members)	
Edge-AI deployment integration		Yes (all members)	
Overall system testing		Yes (all members)	

5. USER REQUIREMENTS

5.1 User and Stakeholder Requirement Identification

Indirectly, through the group's shared decision-support dashboard in Component 4, the primary beneficiaries of the output of Component 3 will be agricultural field officers and plantation managers, who will use the severity scores, categories and trends of progression. The following requirements were found at first during the projects statement to have severity labelling verified by experts and to have the severity categorisation and identification of progression trends. These requirements will be further checked by having structured discussions with the agricultural officers involved at the collaborating private Cavendish banana plantation, this will be documented as it happens in Appendix 1 and Appendix 2.

5.2 Identified User Requirements

Users need a quantitative, precise output that makes sense for them - both as a numeric score that they can use to compare themselves to others and as a Low, Moderate, High, or Critical category for immediate communication to the field. Users need the progression trend to be distinct from the current severity trend because a plant may be presently either in the Moderate category or the Worsening category, and this distinction is important if the user needs to know whether the severity is increasing or decreasing, and therefore whether the situation is urgent and should be treated or not. Users must be able to track how the severity assessment was performed, back to an underlying image and, if available, to the environment in which the image was created, to enable expert review and confidence of the automated output.

5.3 Functional Requirements

Table 5 — Functional Requirements

ID	Functional Requirement
FR1	The component shall receive a disease classified leaf image and metadata (plant identifier, zone, date) from component 1.

FR2	The component should divide an input leaf image into leaf and lesion regions using a U-Net model.
FR3	The component should determine the percentage of lesion area to leaf area (Scope) from the Segmentation Results.
FR4	The component shall determine the Impact score for the disease detected based on an existing or proposed severity standard based on the disease.
FR5	The component should obtain the environmental risk (Urgency) score pertaining to the zone and date from Component 2.
FR6	The component should calculate a severity score based on the combination of Impact, Urgency and Scope, defined by the validated fusion method.
FR7	The component shall categorise the severity score as Low, Moderate, High or Critical.
FR8	Each severity observation should be stored with the plant/leaf identifier and the date of observation to track over time.
FR9	The component shall use the current severity indicator score for the plant to determine whether it has gotten worse, stayed the same, or improved from its previous recorded score(s) and classify it as Worsening, Stable, or Improving respectively.
FR10	If there are adequate repeated observations, the component shall forecast the severity in the short term.
FR11	The severity score, category, lesion-area percentage, trend and forecast shall be sent to Component 4 if available, from the component.

5.4 Non-Functional Requirements

Table 6 —Non - Functional Requirements

Requirement	Measurable / Testable Target
Accuracy	Severity classification will obtain a macro-F1 score higher than the baseline that is derived from the rule. The macro-F1 score of non - Severity classification will be higher than the baseline score derived from the rule on the held-out plant-level test set.
Reliability	When there is not enough temporal data to classify a progression, the component shall report low confidence instead of a silent failure.
Response Time	The severity score for a single image should be completed within a time budget that is compatible with the group's Edge-AI real-time inference requirement, in cooperation with Component 1's deployment benchmark.
Usability	Severity categories and progression trend shall be displayed in plain language without the need of machine learning specific interpretation by field user.
Scalability	The component should be able to handle severity assessment for the four target diseases without having to redesign the architecture if a fifth disease should be added in the future.

Maintainability	Sub-models of Segmentation, scoring and forecasting shall be developed and available for deployment as stand-alone modules for independent retraining.
Traceability	All severities will be stored with the source image reference, plant/leaf identifier, observation date, and model version.
Data Privacy	Personally identifiable farmer/landowner data shall not be stored or included in the images and metadata, and the identity of the collaborating plantation shall not be revealed without its explicit consent.

6. Commercialization Plan

6.1 Target Market and Customer Persona

The main target market will be commercial and estate scale Cavendish banana producers in Sri Lanka, and agricultural extension services catering to small holder producers. The main customer persona is a plantation field officer or agronomist who is currently doing manual subjective severity assessment, and would like to have a more consistent, concise and quantifiable indicator of severity to help prioritize which areas to treat.

6.2 Value Proposition

The ability to convert subjective, manual visual severity assessments to a quantitative, repeatable severity score and category, along with environmental context, and a progression indicator that is not possible to get at scale from disease presence alone, is what Component 3 offers to support earlier and better-targeted treatment decisions.

6.3 Revenue Model

The severity assessment capability of Component 3 would form part of the shared group system, which would be sold as a business-to-business (B2B) licence or subscription to plantation operators and charged based on the size of the plantation and/or number of zones being monitored, in line with the group's overall commercialisation strategy for developing an integrated Edge-AI system.

6.4 Cost Estimation

Table 7 — Cost Estimation

Cost Category	Main Expenses
Development	Segmentation model development, feature extraction, XGBoost/GRU training pipelines
Data and Annotation	Expert annotation time, annotation-tool subscription, field-image collection support
Cloud/Compute	GPU compute for U-Net and GRU training, cloud storage for annotated datasets

Maintenance	Periodic retraining as new severity-labelled data accumulates, model monitoring
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6.5 Pricing Strategy

As in the rest of the group's system, pricing would be based on plantation size and monitoring frequency – and the severity-and-progression feature would be a higher-level decision-support aspect of the system, beyond simple disease detection.

6.6 Competitive Advantage

The competitive advantage of Component 3 is its ability to integrate the image-based severity, environmental context and the tracking of temporal progression into the single output, something not observed in existing banana disease research or, if applicable, existing commercial tools for banana disease detection, which primarily perform classification.

6.7 Intellectual Property Considerations

The segmentation and severity-scoring models, proposed severity proxies for Cordana and BBTv, and any resulting dataset that is annotated to include the proposed models, will be governed by SLIIT's research intellectual property policy, and the terms of the data-sharing agreement negotiated with the collaborating Cavendish banana plantation. Original raw images from the field and any metadata that identifies the plantations is retained with the collaborating plantation and is treated as confidential through this proposal.

7.BUDGET AND JUSTIFICATION

The individual research costs for Component 3 are budgeted in the following budget estimate. Figures are estimates for supervisors' approval and are stated as estimated or subject to quotation where the exact market prices have not been independently verified, and will be adjusted when annotation volume and compute requirements have been verified during the data-collection phase.

Table 8 — Budget and Justification

Item	Description	Quantity	Unit Cost (LKR, est.)	Estimated Cost (LKR)	Justification
Cloud GPU compute	Training compute for U-Net and GRU experimentation	~40 hours	375 (est.)	15,000	Segmentation and temporal-forecasting model training and experimentation
Annotation tool subscription	CVAT/Roboflow individual share	1 subscription period	6,000 (est.)	6,000	Expert lesion/leaf mask and severity annotation

Field data-collection support	Travel to the collaborating plantation for image collection and expert-validation sessions	Multiple visits	N/A	8,000	Site visits required for field imagery and expert severity validation
Cloud storage	Storage for the expert-annotated severity/segmentation dataset	12 months	~415/month (est.)	5,000	Storage for annotated image dataset and model artifacts
Printing and documentation	Proposal, progress reports, and documentation printing	As required	N/A	3,000	Required academic documentation

8. WORK BREAKDOWN STRUCTURE (WBS)

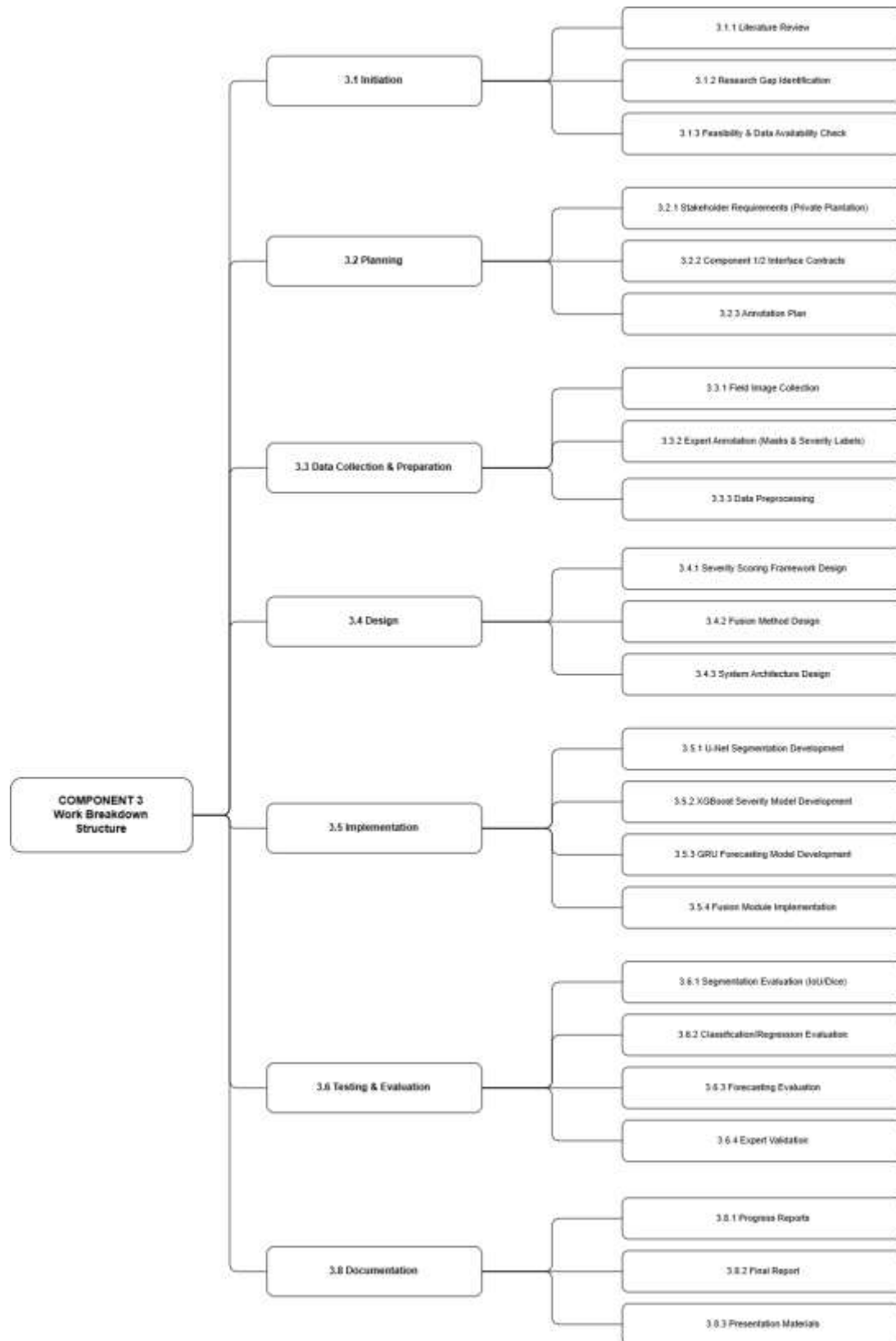


Figure 3 Work Breakdown Structure

9. GANTT CHART

GANTT CHART - COMPONENT 3 DEVELOPMENT TIMELINE (SEP 2026 - JUN 2027)

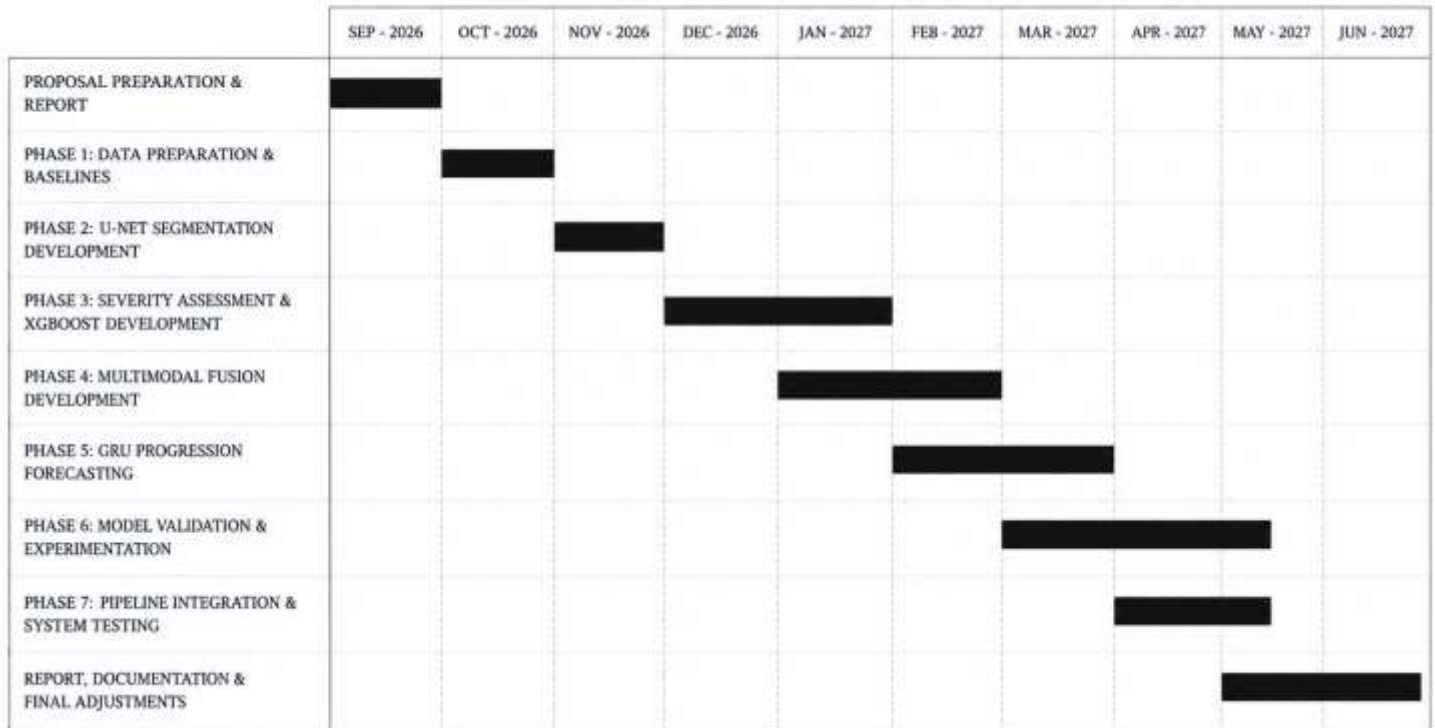


Figure 4 Gantt chart

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11. APPENDICES

APPENDIX 1

To be populated with notes from discussions with agricultural officers at the collaborating private Cavendish banana plantation regarding current manual severity-assessment practice; photographs or examples of field severity-assessment forms currently in use, where available; and preliminary sample images collected during site visits.

APPENDIX 2 - Mandatory Proposal Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
June 2026	No validated severity scale exists for Cordana/BBTV	Literature search across banana pathology sources found no Cordana/BBTV quantitative scale	Apply Sigatoka scale directly vs. propose disease-specific proxy	Propose disease-specific proxies, flagged as unvalidated	[to be recorded]
June 2026	Impact x Urgency x Scope formula has no direct literature precedent	Searched plant-pathology and general risk-assessment literature	Retain as-is vs. redesign vs. critique-and-retain-conditionally	Retain as proposed prototype; critically evaluate against expert rankings	[to be recorded]
June 2026	Field site identity should not be disclosed in the proposal	Consideration of plantation operational privacy	Name the estate directly vs. refer to it generically	Refer to the site as a selected private Cavendish banana plantation throughout	[to be recorded]

Table 8 appendix 02

APPENDIX 3 — AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
Claude (Anthropic)	Literature search assistance	Suggested relevant papers for Severity assessment, Forecasting	Each suggested paper was individually located and read by me. Only verified papers were cited.	I read all papers and drew my own suggestions.

Table 9 appendix 03

Declaration

I accept full responsibility for the accuracy, originality, validity of references, research decisions, methodology, technical choices, interpretation of evidence, and the ability to defend every part of this proposal.